

MOHAMED “OSCAR” GHANEM

 Oschart |  maghanem |  Publications
Saarbrücken, Germany | m.ghanem.mg16@gmail.com

FIRST-AUTHOR PUBLICATIONS

- “The Shape of a Program: Path Signatures for Trace-to-Program Induction.” *NeurIPS 2026*.
- “Flowing Through States: ODE Regularization for Reinforcement Learning.” *ICLR 2026*.
- “Learning Better Representations From Less Data For Propositional Satisfiability.” *NeurIPS 2024 (Spotlight)*.
- “FLoBC: A Decentralized Blockchain-Based Federated Learning Framework.” *IEEE BCCA 2022*.

EMPLOYMENT

Doctoral Researcher

February 2024 – Present

CISPA Helmholtz Center

- Lead research on learning for formal reasoning and control, spanning neuro-symbolic SAT solving and dynamics-regularized reinforcement learning; first-author papers at NeurIPS 2024 (Spotlight) and ICLR 2026.
- Designed a certificate-based framework vastly performing prediction-based learning for SAT solving (NeurIPS 2024).
- Proposed an ODE-based state regularizer that improves sample efficiency and stability of RL agents (ICLR 2026).
- Tutored the Neuro-Symbolic Computing 2024 seminar; reviewer for NeurIPS, ICLR & ICML (2025–2026).

Machine Learning Engineer

October 2021 – December 2023

Spree3D

- Developed an automated evaluation suite for Virtual Try-On models with neural KPIs (e.g., structural similarity).
- Built an end-to-end deep learning pipeline for volumetric 3D hair generation from portrait images.
- Implemented and tweaked a state-of-the-art video segmentation model that served most of our machine learning pipelines.
- Undertook many large-scale GAN-based dataset generation processes producing terabytes of training data.
- Collaborated on developing a deep learning pipeline to produce 3D head geometry from portrait images.

Research Assistant

June 2022 – April 2023

Max-Planck Institute: MPI-SWS

- Developed a PyTorch implementation for a meta-learning technique that outperformed most benchmarks.
- Built a unifying Python library for all prominent reward-shaping techniques following a **stable-baselines3** style.

Automated Reasoning Tutor

November 2021 – February 2022

Max-Planck Institute: MPI-INF

- Conducted weekly tutorials and graded submissions on the foundations of automated theorem provers.

Data Scientist

July 2021 – October 2021

Turing.com

- Developed an adaptive time-series model for predicting ad-campaign revenue that remains in production use today.
- Created numerous automated analytics reports for product and marketing teams.

Software Engineer

July 2019 – October 2020

Nabta Health Ltd.

- Created numerous React Native components corresponding to four of the main app’s primary features.
- Built NFC temperature sensor drivers on both native platforms (Android Java & iOS Obj-C).
- Implemented several REST APIs essential for content-fetching and database synchronization.

EDUCATION

Technical University of Munich

February 2024 – Present

Munich, Germany

- **PhD in Computer Science:** Machine Learning, *expected 2028*. **Advisor:** Prof. Bernd Finkbeiner.

Saarland University
Saarbrücken, Germany

October 2021 – January 2024

- **M.Sc. in Computer Science**, GPA: **1.3** (German scale, 1.0 = best)
- **Graduate Coursework**: Reinforcement Learning, Machine Learning in Cybersecurity, Neuro-Symbolic Computing, Automated Reasoning, Security, Machine Learning for Program Analysis, Automated Debugging
- Completed two **Research Immersion Labs** with Prof. Adish Singla and Prof. Bernd Finkbeiner.

The American University in Cairo
Cairo, Egypt

September 2016 – June 2021

- **B.Sc. in Computer Engineering**, GPA: **3.97/4.0** (American scale)
- **B.Sc. Thesis**: *FLoBC: A Decentralized Blockchain-Based Federated Learning Framework*
 - This project offers a collaborative framework for training ML models without sharing user-specific training data.
- **Minor in Mathematics**, GPA: **4.0/4.0**
- **Mathematics Thesis**: *On Theoretical Complexity and Boolean Satisfiability*

ACADEMIC SERVICE

- **Reviewer**: NeurIPS 2025 & 2026, ICLR 2026, ICML 2026 (*Silver Reviewer*).
- **Subreviewer**: AAAI 2025.

ADDITIONAL EXPERIENCE AND AWARDS

- **2nd Place Winner**: Dell's C(4F)2 Artificial Intelligence Hackathon 2021 (students track).
- **Contestant**: ACM Arab and African Collegiate Programming Contest (**ACPC**) in the **2018 & 2019** versions.
- **Finalist**: ACM Egyptian Collegiate Programming Contest (**ECPC**) in the **2018 & 2019** versions.
- **Collegiate Award Holder**: Full-tuition PSSF scholarship (115k USD) for outstanding academic performance.

LANGUAGES AND TECHNOLOGIES

- **Languages**: Python, Lean, C/C++, CUDA C, Rust
- **Tools/Frameworks**: PyTorch, JAX, Docker